What Deception Probes Read and How Evaluations of Them Fail

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Abstract

Linear probes on the residual activations of large language models are the leading proposal for monitoring deceptive generation. We ask what such probes read, where that signal is computed, and whether the standard experiments used to answer these questions are themselves sound. Our answers, from a fully pre-specified pipeline on nine open models: the probes read truth-consistency rather than deception; the computation behind that signal is not identified across models under this pipeline; and the standard evaluations returned wrong answers at nearly every step until controlled. The testbed is instructed lying in a forced-choice Yes/No format, a measurable proxy for deception, and every claim is scoped to it. On the four models with per-layer ablation curves (Qwen2.5-0.5B/1.5B/3B, Gemma-2-2b), deception-specific damage is band-localized at 1.5B, multi-site at Gemma-2b, present locally but absorbed at the deployed readout at 3B, and absent at 0.5B; clean patching restores 0.94–0.98 of what ablation removes wherever a critical layer exists. The standard steering attack, subtracting the probe direction at the readout layer, is a translation whose effect on AUC is analytically and measurably zero: it defeats the thresholded monitor operationally while provably leaving the representation intact. Out of format, instructed-transfer evaluation collapses: naive transfer AUCs span 0.030–1.000 across nine models, denial bias leaves seven of nine unidentifiable, and in the two identifiable models the instruction coefficient is roughly four times a sign-inconsistent deception effect. We distill six controls into an evaluation protocol; code, data, splits, and a regenerable results file are released.

1 Introduction

Linear probes on residual activations are the leading proposal for monitoring deception in language models: they are cheap, they report near-ceiling accuracy on their training distributions, and they are already being evaluated for deployment. Three questions underneath that proposal remain open. What do such probes actually read? Where is that signal computed? And are the standard experiments used to answer the first two questions, point interventions at selected components and instructed-lying transfer tests, themselves sound? We answer all three on deliberately small models (0.5B–3B parameters, two families), trading construct realism for full causal access, under a pipeline in which every design choice was pre-specified and every number regenerates from a released results file. Throughout, deception means instructed lying in a forced-choice format: the model asserts a statement its own representations encode as false. This is the measurable proxy, not spontaneous or incentivized misrepresentation, and §5 shows that even the proxy partially collapses into answer polarity for most models; the identifiability gate exists to detect exactly that collapse.

A single measurement illustrates the problem. Ablating Qwen-3B's critical MLP drops a probe's accuracy from 0.950 to 0.719, a fall of 0.231 that reads unambiguously as a necessary component. The same ablation moves the AUC from 0.992 to 0.979 (Table 1): the ranking the probe reads is intact, and what moved was the decision threshold. The divergence extends to significance. Under balanced accuracy the paired specificity asymmetry at this layer is -0.212 with a bootstrap interval excluding zero; under AUC the deception and truth drops are 0.013 and 0.017, equal and negligible. One metric manufactures a mechanism with a confidence interval attached; the other shows that nothing happened.

Table 1: De-calibration exhibit for Qwen-3B: the same ablation under two metrics.

Metric (Qwen-3B, critical MLP)	Clean	Ablated	Difference
Balanced accuracy	0.950	0.719	-0.231
AUC	0.992	0.979	-0.013
Paired asymmetry, balanced accuracy	-0.212 $[-0.275, -0.150]$ (significant)		
Paired asymmetry, AUC	$+0.004$ (deception drop 0.013, truth drop 0.017)		

Contributions.

1. **Causal topology, curve-adjudicated (§4):** four models, four shapes under one pipeline (banded, multi-site, locally present but absorbed, absent); conduit sufficiency 0.94–0.98; no scale law (App. G).
2. **A methods result (§4, §5):** magnitude-based point interventions misidentify the mechanism; only per-layer curves with label decomposition adjudicate.
3. **Transfer evaluation fails measurably (§5):** behavioral degeneracy, naive statistics spanning $[0, 1]$, and instruction-dominated regressions; each failure caught by a specific control, distilled into a six-step protocol.

Each finding exists in its present form because an earlier version of it was an artifact of a default measurement choice: a recall metric read our strongest necessity effect as an ablation that *improved* detection, and balanced accuracy manufactured the 3B pseudo-mechanism of Table 1. App. F catalogues all six reversals; the protocol of §5 is the set of controls that caught them.

2 Related work

Deception and lie-detection probes. Azaria and Mitchell (2023) train a three-layer feedforward classifier on hidden-layer activations to predict statement truth, reporting 71–83% under topic-held-out evaluation; Burns et al. (2022) probe truth without supervision. Goldowsky-Dill et al. (2025) report near-ceiling detection (AUROC 0.96–0.999) on contrasting honesty instructions, and observe two effects they do not pursue: sensitivity to whether the scenario relates to deception rather than whether the response is deceptive, and recall collapsing when the readout position moves. Our results are the quantified form of both observations; our contribution is not that these probes are fragile, which that work notes, but that the evaluations used to certify them return uninterpretable numbers unless these effects are controlled for. Pacchiardi et al. (2023) detect lying from follow-up behavior; representation engineering generalizes the recipe (Zou et al., 2023). Levinstein and Herrmann (2023) are the closest precedent for our transfer results: reconstructing probes of this family on LLaMA-30B, they find probes trained on positive statements score below chance on negations, and argue such probes learn a correlate of truth on the training set rather than truth itself. Our claim is one level up: we take the truth-consistency signal as causally real, having localized, ablated, and restored it, and ask instead whether it is deception. Their negation result is untested in our setting and recorded as a limitation.

Geometry of truth. Marks and Tegmark (2023) characterize low-dimensional truth subspaces and intervene causally along probe directions, evaluated out of distribution, as does inference-time intervention (Li et al., 2023); they also report that difference-in-means probes match logistic regression on accuracy while being substantially more causally implicated, the basis of our probe-family limitation. Kumar (2026) pressure-tests deception-probe geometry observationally across Gemma-3 (1B–27B), rejecting a single-direction account. Two findings there bear directly on ours. Stratifying by whether the model actually lied or merely operated under a lie instruction, that work obtains identical AUROC for both strata and concludes the probe detects the deceptive prompt regime rather than the downstream success of deception, the conclusion our de-confound statistic and polarity-controlled regression reach by a different route in §5. It further notes that a definitive test would require deception data with no system-prompt manipulation; our causal pipeline is teacher-forced and contains no instruction at all, so the in-format results of §4 are not subject to that confound, while our out-of-format tests are and are reported as such.

Probing methodology. Our protocol extends a tradition of probing skepticism, from control tasks and amnesic probing to the interpretability-illusion line (Hewitt and Liang, 2019; Elazar et al., 2021; Bolukbasi et al., 2021; Belinkov, 2021); we contribute the transfer-evaluation analogues: degeneracy gates, instruction-leak de-confounds, and position and metric controls.

Causal interventions and patching metrics. We use causal tracing (Meng et al., 2022) and path-level analysis (Wang et al., 2022). The closest prior work to our metric thesis is the activation-patching methodology line: Zhang and Nanda (2024) show probability as a patching metric fails to detect negative components when corruption drives the correct-token probability near zero, and Heimersheim and Nanda (2024) warn that discrete metrics can crown one contributor among many because it happens to cross a threshold. Table 1 is a measured instance of that failure. We differ from this line in three ways. The object differs: their metrics evaluate a model's next-token distribution under a patch, in service of a claim about a circuit, while ours evaluates a deployed classifier's readout, in service of a claim about a monitor. The failure modes differ: behavioral degeneracy, instruction leak, and readout-position mismatch have no patching analogue, since they arise only when the object under evaluation is a trained detector asked to generalize across conditions. The evidential status differs: we report a pre-specified measurement with a falsification rule across nine models, in which six conclusions were reversed by the controls before publication. The deployed-readout compensation we observe connects to self-repair (McGrath et al., 2023; Rushing and Nanda, 2024); §6 treats what that literature explains about our results and what it does not.

Evasion. Learned attacks defeat latent-space monitors (Bailey et al., 2024; Gupta and Jenner, 2025; McGuinness et al., 2025), with token-level monitors more vulnerable than pooled ones (Gupta and Jenner, 2025) and ensembles more resilient than single directions (McGuinness et al., 2025). We show measurement breaks before any adversary arrives.

3 Setup

Pipeline. We use 400 true/false statements (200/200, each at most twelve words), separated into statement-grouped splits: both members of a statement pair fall on one side of every split. The readout layer is selected by nested selection (layer accuracy averaged over 10 validation splits, earliest layer within 0.02 of the peak, held-out test fold never used). One probe per model: fit once on the combined train and validation folds at the selected layer, then frozen and reused unchanged in every analysis of §4 and §5, so the object localized and the object transferred are the same classifier. Critical MLPs are pre-specified as the layer whose ablation most reduces that probe family's continuous score, read at the ablated layer, on training-fold deceptive items. Throughout, *pre-specified* means fixed before measurement and published with the code; *pre-registered* is reserved for the addenda committed before their data existed. All causal effects are reported threshold-free (AUC), with balanced accuracy in App. F; grouped bootstrap over statements for every interval. One device and precision (A100, fp16); every number regenerates from the released results file; environment manifests are in the anonymized repository.

Dataset. The primary corpus was assembled in two batches (200 covering geography, science, history, elementary mathematics; 200 constructed as minimal true/false pairs over the same domains), drafted with LLM assistance and reviewed by the authors for factual accuracy. Eight statements recur across batches in near-identical form; two of those pairs straddle the train/test boundary, bounding test-accuracy inflation at 1.25 percentage points (App. C). A secondary set of 100 statements (50/50) was written by a non-author with assistance from a different model family; it tests generalization across author, model family, and collection pipeline. A replication on an external corpus (Azaria and Mitchell, 2023) is pre-registered at probe level only and not yet run; no claim here depends on it.

Labels. Two pipelines with different label provenance. In the *causal* pipeline (all ablation, patching, and specificity results), the answer token is teacher-forced: each statement is presented with an appended Yes/No answer and activations are read at that token, so deception, truth, and polarity labels are defined by construction. In the *behavioral* pipeline (§5), the model generates freely; non-parseable generations go to a separate incoherent bucket rather than defaulting to a class (empty in every condition), and ambiguous generations are dropped at filtering (n_{kept} 461–588 of 800). Where

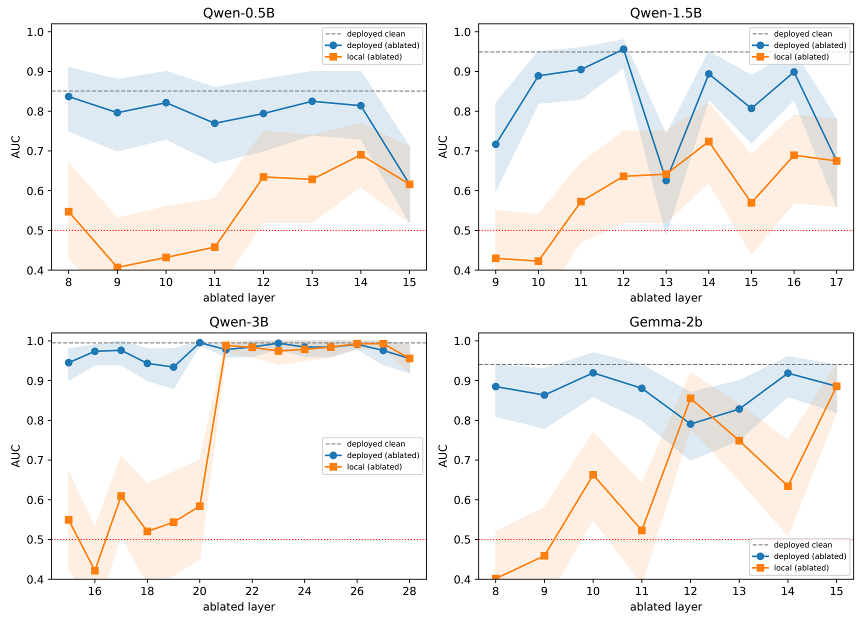

Figure 1: Per-layer ablation damage across four models: one intervention, four pictures. Dashed line: deployed AUC with no ablation. At Qwen-0.5B the deployed reading falls only at L15, the readout layer itself, so the ablation is circular. At Qwen-1.5B specific layers cut deployed strength sharply, deepest at L13. At Qwen-3B the deployed reading holds above 0.93 everywhere. At Gemma-2b it dips across L11–L13.

a model lies purely by denial, polarity and response direction are near-collinear by construction; this is the denial-bias phenomenon of §5, not a labeling artifact, and such models are gated out of the transfer regression.

Probe strength. Ten-seed accuracy at the selected layer is 0.698, 0.866, 0.929 and 0.836 for 0.5B, 1.5B, 3B and Gemma-2b, transferring to 0.750, 0.920, 0.960 and 0.955 on the secondary set (App. C); the secondary statements are simpler (mean 4.9 vs. 6.8 words), which is why transfer exceeds in-distribution accuracy for three of four models. An answer-word leak control, a design-level check independent of the model, sits at 0.456 ± 0.026 across ten seeds, so answer wording alone carries no exploitable signal.

4 Causal topology

We ablate every MLP in each model's recovery band and read deception, truth, and polarity probes threshold-free, at both the ablated layer and the deployed monitor (Fig. 1, Table 2). This gives a per-layer view of where deception-specific damage occurs, rather than a single summary at one selected component. The curve analysis covers the four models where per-layer ablation was affordable; the nine-model census in §5 uses single-readout measurements. All ablations are zero-ablations of the MLP output; this is a noising intervention, and §6 states what that bounds.

Table 2: Causal summary per model: critical layer, necessity AUC at local and deployed readouts (ablated in parentheses), restoration fraction, max deployed damage, topology.

Model	Crit	Local nec.	Deployed nec.	Restoration [CI] (n)	Max dep. damage	Topology
Qwen-0.5B	L15	0.851 (0.616)	0.851 (0.616)	0.671 [0.539, 1.208] (77)	0.235 (L15*); 0.081 else	Absent
Qwen-1.5B	L16	0.924 (0.689)	0.949 (0.899)	0.970 [0.959, 0.980] (80)	0.324 (L13)	Banded (L15–17)
Qwen-3B	L24	0.992 (0.979)	0.995 (0.978)	0.945 [0.834, 1.011] (49)	0.061 (L19)	Local, absorbed
Gemma-2b	L9	0.728 (0.459)	0.941 (0.864)	0.980 [0.960, 0.992] (65)	0.150 (L12)	Multi-site (L9,11,13,14)

*L15 is the deployed layer for Qwen-0.5B; the self-ablation value is not citable as necessity.

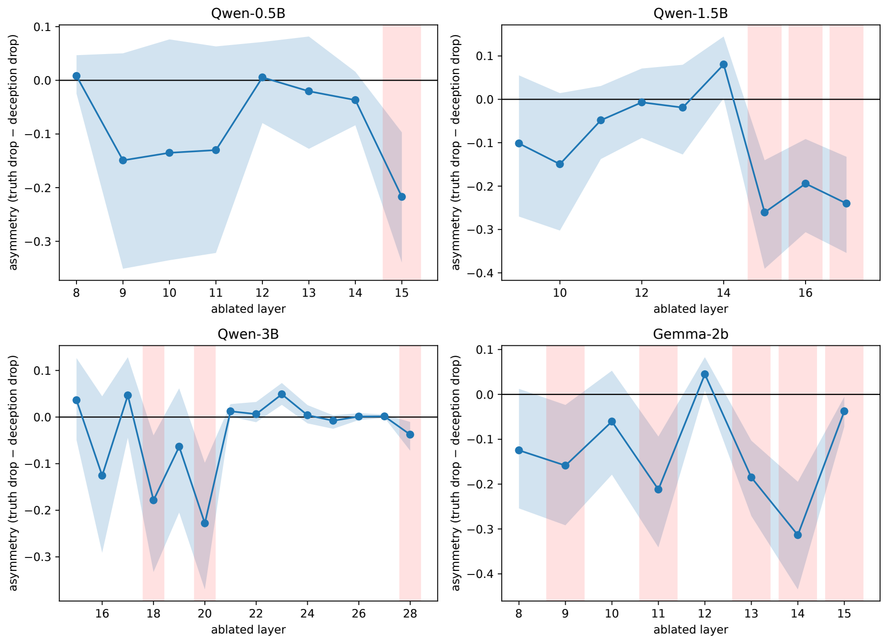

Figure 2: Paired asymmetry (truth drop − deception drop) per ablated layer; shaded band shows the grouped-bootstrap CI, red spans mark significant deception-specific damage. The shape differs in every model.

Because the answer token is teacher-forced, polarity survival is a stream-integrity floor: ablation leaves token-level information fully intact, so no ablated component is a generic bottleneck, and the mechanistic claim is carried by the *differential* damage between deception and truth readouts (paired asymmetry), not by polarity itself. Across all 43 ablation conditions (39 per-layer cells, 8/9/14/8 across the four models, plus four paraphrase replications), polarity AUC remains 1.000 without exception. AUC specificity at the critical layer is tabulated in App. F (Table 18), directly alongside its balanced-accuracy counterpart (Fig. 3).

Deception-specific damage takes a different shape in every model (Fig. 2). At Qwen-1.5B it occupies a contiguous three-layer band (L15–L17; asymmetries -0.261, -0.194, -0.240, all CIs excluding zero). At Gemma-2b it is multi-site: four interleaved layers (L9, L11, L13, L14) reach significance, maximal at L14 (-0.314), while the intervening L12 is significantly *truth*-specific, so the sign alternates between adjacent layers. At Qwen-3B the picture is different again, and it is why we do not describe that model as simply lacking structure: significant deception-specific damage *is* present locally, at L18 (-0.178) and L20 (-0.228), while the deployed monitor at those same layers reads 0.943 and 0.995, essentially unmoved. Local damage occurs; the deployed readout absorbs it (Fig. 5, App. C). At Qwen-0.5B nothing is localizable: the probe is weak (0.698) and selection degenerates to the readout layer itself. Cell significance is corrected for multiplicity within each model's family by Benjamini–Hochberg at $q = 0.05$, on two-sided bootstrap p-values computed from retained draws (App. H; draws released). Every layer identity cited above survives: the 1.5B band at all three layers, all of Gemma-2b's sites, and both of Qwen-3B's local sites. Across the 39 cells correction removes exactly one, a $+0.012$ truth-specific effect at 3B's L21 that carries no claim. Four further cells at Gemma-2b and Qwen-3B reach significance and are set aside for reasons other than multiplicity: Gemma-2b's L12 and Qwen-3B's L23 are truth-specific rather than deception- specific, and Gemma-2b's L15 and Qwen-3B's L28 are those models' own readout layers, so their ablation is circular. Full accounting in App. H, Table 21.

The curves also show how selection rules fare, including our own. Our pre-specified rule picks the layer whose ablation most reduces the probe score *at the ablated layer*. At 1.5B it selected L16, inside the significant band, and the choice survives; but the single most damaging ablation at the *deployed* readout is L13 (0.949 → 0.625), a layer with no deception specificity (asymmetry -0.019), and any rule keyed to deployed damage would crown it the mechanism. At Gemma-2b the rule

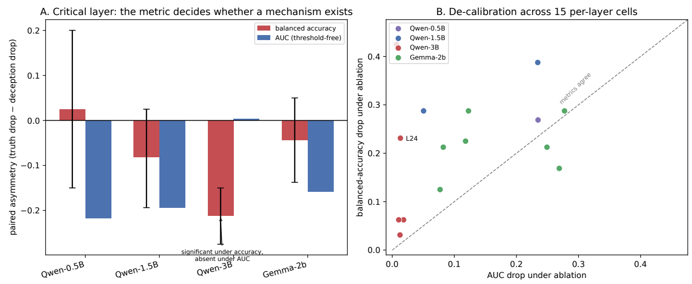

Figure 3: Left: paired asymmetry at each model's critical layer under balanced accuracy (with bootstrap CI) and under AUC; at Qwen- 3B the accuracy metric reports a significant mechanism where the threshold-free metric reports none. Right: balanced-accuracy drop against AUC drop per recorded cell; the diagonal is agreement, height above it is de-calibration, and Qwen-3B's critical layer (labelled) is the extreme case.

selected L9, a significant site but not the maximal one (L14), and the largest deployed damage (L12) is truth-specific. At 0.5B the rule degenerated to the readout itself. One pre-specified rule, three different outcomes across models: magnitude and mechanism dissociate in both directions, and only the curves adjudicate.

Sufficiency. Necessity establishes that a component matters; sufficiency asks whether it carries the signal. We corrupt a run by subtracting the unit probe direction upstream, then compare four conditions: clean; steered; steered with the critical MLP also ablated (the floor); and patched, restoring the MLP's clean output into the steered run. Restoration is the fraction of the floor-to-clean gap the patch recovers. Where a critical layer is identifiable it behaves as a conduit: restoration 0.970 [0.959, 0.980] at 1.5B, 0.980 [0.960, 0.992] at Gemma-2b, and 0.945 [0.834, 1.011] at 3B, the last on the 61% of items where the ratio is defined ($n = 49$); intervals exceeding 1 reflect ratio-estimator instability, not super-restoration. Qwen-0.5B gives 0.671 with a wide interval and is not leaned on. The contrasting statistic, whether the patch undoes the upstream corruption itself, sits near zero (-0.106, -0.063, $+0.119$), as a conduit requires (App. C).

No scale law. Maximum deployed damage orders 0.32 / 0.15 / 0.06 across 1.5B / Gemma-2b / 3B, with 0.5B's 0.24 arising solely from readout self-ablation. A pre-specified falsification test found the ordering non-monotonic in parameter count and depth; the trend was retired via a logged deviation from that rule (App. G).

Evasion separates de-calibration from destruction. Subtracting $\alpha\hat{\theta}$ at the readout layer collapses thresholded detection (below 0.5 by $\alpha = 2$ in all models) while leaving behavior untouched (App. C). But at the readout layer the intervention is a translation, so its effect on a linear probe is exact: every decision score shifts by the same constant $-\alpha\|w\|$, rank order is preserved, and AUC is unchanged; we measure that change at 0.000000 in all four models. Since deployed monitors are thresholded, this attack defeats the monitor operationally, and the decomposition says what was defeated: the calibration, not the signal. Injected *upstream*, at half the readout depth, the same direction is a different intervention: collapsing detection there requires 3.3–7.5× more strength ($\alpha = 10/15/20/30$ against $2/4/6/4$), the representation genuinely degrades, and yet a probe refit on the steered activations recovers 0.94–0.99 of clean AUC at 1.5B, 3B and Gemma-2b; only at 0.5B does the signal fail to return (0.564 on the canonical split, 0.537 ± 0.029 over ten seeds). At 3B the frozen probe's AUC moves only 0.998 to 0.984 while detection falls to 0.087, so even the upstream attack is de-calibrating there. This test was pre-registered after the main results were fixed (per-model numbers and one registration ambiguity, App. C).

4.1 Format stability of the address and fingerprint

We re-ran the entire pipeline under three phrasings (original, true-or-false, claim), each processed independently from probe training through critical-MLP identification. In both models with localizable structure the critical layer is re-selected identically under all three formats (1.5B: L16/L16/L16; Gemma-2b: L9/L9/L9), probe accuracy remains strong (0.838–0.912), and the specificity fingerprint is format stable: deception AUC drops 0.15–0.30 at the critical layer while truth barely moves and polarity holds at 1.00 (Table 8, App. C). Steering resistance, by contrast, is format dependent: collapse-α grows from 4 to 16–24 under paraphrase. Stability across *formats* does not imply stability across *corpora*; §6 states that as an open question.

5 Why instructed-transfer evaluation fails

A monitor is only useful if it survives contact with a setting it did not see. The standard test is instructed transfer: train on contrastive honest/deceptive pairs, then evaluate on a model told to lie. We ran that test on all nine models and obtained nine confident answers spanning nearly the entire range of the metric (Table 3). Each control below was added because the preceding one was insufficient, and each changed a conclusion we had already drawn.

The de-confound statistic is the frozen probe's AUC at separating honest-condition responses from non-compliant lie-condition responses (told to lie, answered truthfully). Both groups contain only truthful answers, so separation can only come from the instruction context; 0.994 and 0.002 are both leaks, one read forward and one inverted. We flag $|x - 0.5| > 0.15$; all nine models exceed it. Because non-compliant responses are a selected subset, under denial bias the two groups can differ in statement composition, so we recompute the statistic with truth and polarity held fixed by stratification (exploratory, App. D): eight of nine still leak, matching moves the statistic *away* from 0.5 in four models, the two identifiable models are essentially unchanged (0.991, 0.984), and only Qwen-1.5B (0.365) moves marginally inside the band. Composition is a real confound and it does not explain the leak.

Table 3: Transfer census: naive statistics span the full interval; degeneracy leaves 7/9 unidentifiable.

Model	Pos. control	Transfer AUC	De-confound	Denial asym.	Verdict
Qwen-0.5B	0.826	0.030	0.019	$+1.00$	Degenerate
Qwen-1.5B	0.948	0.499	0.176	$+1.00$	Degenerate
Qwen-3B	0.998	0.998	0.913	$+1.00$	Degenerate
Qwen-7B	0.998	0.992	0.942	$+1.00$	Degenerate
Qwen-14B	0.999	1.000	0.716	$+1.00$	Degenerate
Gemma-2b	0.947	1.000	0.994	-0.16	Identifiable
Gemma-9b	0.996	0.030	0.002	-0.69	Degenerate
Llama-3.2-3B	0.968	0.920	0.982	$+0.94$	Degenerate
Llama-3.1-8B	0.994	0.878	0.980	$+0.365$	Identifiable

Three of nine models meet or exceed their own positive-control ceiling (Gemma-2b by 0.053; Qwen-14B and Qwen-3B within noise of a ceiling that is itself saturated), a warning about ceiling-based checks (App. C, Table 9).

(a) Denial bias. Instructing each model to lie about all 400 statements, all five Qwen scales lie only by denying truths (deny-true 100%, affirm-false 0%, scale-invariant 0.5B–14B); Llama-3.2-3B is nearly as one-sided ($+0.94$); Llama-3.1-8B is partially bidirectional ($+0.365$); Gemma-2b is the most bidirectional (-0.16); Gemma-9b affirms every false statement (-0.69) (Fig. 6, App. C). The consequence is not noise but non-identifiability: where deception and answer direction are near-collinear, no probe can separate them, and a transfer result cannot be interpreted regardless of its value. Gemma-9b is excluded by a pre-specified minority-cell floor of 0.20 (observed 0.155); any threshold between 0.155 and 0.318 yields the same partition (App. D).

(b) The regression. For the two identifiable models we regress standardized probe scores on instruction, truth, polarity, and their interaction (App. D). In this saturated parametrization the deception contrast is $-\beta_{\mathrm{truth}\times\mathrm{polarity}}/2$, since a response is deceptive exactly when answer and truth disagree. Gemma-2b: instruction $+1.617$, deception $+0.403$; Llama-3.1-8B: instruction $+1.468$, deception -0.375 (all $p < 0.001$; condition numbers 9.7, 10.4). The instruction coefficient is roughly four times the deception effect in both models, and the deception effect is sign-inconsistent between them: out of format, the probe transfers as an instruction detector with model-idiosyncratic residual deception loading. An earlier version of this analysis, before layer selection was corrected to nested validation folds, reported a negative deception coefficient in both models; the corrected pipeline flips Llama-3.1-8B's sign (App. F). Further specifications, reported but not cited, are in App. D.

(c) Readout-position sensitivity is itself model dependent. At the pre-specified 0.15 threshold: Gemma-2b's naive de-confound moves from 0.737 (pre-answer) to 0.994 (answer token), sensitive; Llama-3.1-8B moves from 0.988 to 0.980, insensitive, the instruction signal saturating both positions.

(d) Preprocessing (App. E). Under the corrected readout position, all eight basis and scaling cells agree within 0.03; a loader flip in earlier drafts existed only in interaction with the position error.

(e) Metric confound. Worked examples in App. F, headlined by the 3B pseudo-necessity of Table 1 and Fig. 3.

Protocol.

1. Match probe readout position between training and evaluation.
2. Gate transfer tests on behavioral non-degeneracy; exclude, don't average.
3. Require a positive-control ceiling; any transfer number at or above it is likely a confound.
4. Estimate effects with polarity-controlled regression, never from subgroup AUCs.
5. Report causal effects threshold-free (AUC), never as recall on one class.
6. Localize with per-layer curves plus label decomposition, never selected-point interventions.

6 Discussion

These probes read truth-consistency: whether the model's answer matches the fact, rather than whether the model is lying. The signal is real, surviving a change of author at 0.75–0.96, and specific: ablating the layers that carry it damages deception readout more than truth readout. But out of format it behaves as an instruction detector: a probe can tell whether the model was *told* to lie with near-perfect accuracy while barely distinguishing whether it *did*, and the residual deception signal points in opposite directions in the only two models where it can be measured. Any monitor trained on contrast pairs inherits this profile; that an independent observational study reaches the same conclusion on a different model family by stratifying on realized deception (Kumar, 2026) suggests it is a property of the training recipe rather than of our corpus.

The computation behind the signal is not identified across models under this pipeline: four models yield a contiguous band, four scattered sites, a computation whose local damage the deployed readout absorbs, and no localizable structure at all. Claims that a safety-relevant feature lives at a component are therefore scoped to the model, and the pipeline, that measured them.

What the 3B result can and cannot mean. At Qwen-3B, single-MLP ablation produces significant local deception-specific damage at L18 and L20 while the deployed monitor stays above 0.93 everywhere. Two readings survive, and ablation alone cannot separate them: the computation is genuinely distributed, or it is concentrated and compensated downstream. The self-repair literature makes the second reading concrete: compensation restores roughly 70% of an ablated layer's effect at 7B, so more important components are more likely to appear robust to ablation (McGrath et al., 2023; Rushing and Nanda, 2024), and zero-ablation is a noising intervention, which finds every component of a serially dependent computation but only the terminal node of a redundant one (Heimersheim and Nanda, 2024). A denoising complement, multi-site ablation, or path patching would adjudicate; we report the dissociation and decline to name the topology. A second, independent intervention points the same way: under upstream steering strong enough to drive detection to 0.087, this model's frozen-probe AUC falls only from 0.998 to 0.984, the only model of four to meet the de-calibration criterion (App. C). Module ablation and direction subtraction, two unrelated attacks, both leave its readout ordering intact.

That literature also names a candidate mechanism for Table 1: normalization rescaling (Rushing and Nanda, 2024). If ablation shrinks the residual stream by a factor c, a linear probe's scores compress toward the intercept, moving a fixed threshold while preserving rank order. We pre-registered and ran the test (App. C). At the Table 1 cell itself the strict rescaling model is rejected: it predicts worse than the mean in all four models (R^2 -0.81 to -10.2 at the critical layer, local readout). At 3B's deployed readout it fits partially ($R^2 = 0.82$, the registered outcome), and the corroborating correlation check is supported only at 0.5B. The de-calibration is real and the metric argument stands regardless of mechanism, but it is not a simple norm artifact: whatever rotates the representation without destroying its ordering remains unexplained.

Measurability is itself model-dependent: whether a transfer test is identifiable, whether a statistic is stable to readout position, whether a control layer is meaningful, each varied across nine models. An unprotocolled evaluation of a probe monitor returns a confident number with no interpretation attached.

Extensions. Four follow directly. (i) Pre-registered and pending: a probe-level replication on an external corpus (Azaria and Mitchell, 2023), testing whether the *pitfalls* generalize rather than whether our probe does; the registered rule keys on single-split inflation, which our own data clears at one model of four. (ii) A prediction: direction-level ablation should dissolve the topology we observe at 3B; if it does not, our reading of that model is wrong. The upstream-steering result is consistent with the prediction but does not test it, since subtraction shifts a direction rather than projecting it out. (iii) The critical layer is format stable (§4.1); whether it is corpus stable is untested and would separate a property of the model from one of the training distribution. (iv) Whether the signal survives negation and Boolean composition (Levinstein and Herrmann, 2023; Marks and Tegmark, 2023) is the sharpest available test of its generality, and it interacts with denial bias: most models in our census lie only by answering No, so lying in practice is negation shaped, and a probe family with documented failures on negated statements, evaluated on models whose lies are denials, faces a specific and checkable threat to the construct.

7 Limitations

Several of the limitations here were discovered by the controls this paper proposes, which is the strongest available evidence that the controls do something. The four most impactful are: (1) *construct validity*: the task is teacher-forced knowledge inconsistency, not strategic deception, and all claims are scoped accordingly. (2) *Sample size*: only two of nine models are identifiable under the pre-specified gate, and the deception coefficient's sign is inconsistent between them. (3) *Intervention type*: all ablations are zero-ablations, the intervention that perturbs the residual-stream norm most and takes the model furthest off distribution (Heimersheim and Nanda, 2024; Rushing and Nanda, 2024), and one that cannot distinguish a distributed computation from a redundant one (§6). (4) *Untested Boolean structure*: the corpus contains only simple positive assertions, so the negation failure that is the strongest existing evidence against probes of this family (Levinstein and Herrmann, 2023) is untested here. Fifteen further items, held to the same standard, are in App. H, among them: multiplicity across ablation cells, the single lie instruction, model size, module-level ablation, token-level readout, data construction, probe family, the simpler secondary set, positive-control saturation, single task and language, and white-box steering as an upper bound.

8 Conclusion

Deception probes read a real, causally localizable truth-consistency signal, but its computation is not identified across models under this pipeline, and the standard experiments used to evaluate such probes returned wrong answers at many steps until controlled, reversing our own conclusions six times. We distill those controls into a six-step protocol and release a fully regenerable pipeline; we suggest treating any probe-evaluation number produced without them as provisional. The immediate open question is whether probes can be trained to pass the protocol they are evaluated under.

Reproducibility. Code, data, splits, results, and bootstrap draws: `https://github.com/Andresg324/truth-probes`. Every number regenerates from the released results file; any number not in it was not measured. All results come from clean-room runs on one device class and precision (A100, fp16), reproducible for roughly $25–40 in rented GPU time; total project compute, including development and reruns, was under $150. AI assistance was used for corpus drafting (§3), code, and manuscript drafting; all design decisions, analyses, and claims are the authors'.

Responsible use. We find that standard deception-probe evaluations report confident but uninterpretable numbers; the intended effect is to make monitor evaluation more conservative. The residual risk that documenting failure modes informs evasion is low: our steering attack is white-box, uses the probe's own weights, and its in-format form is a threshold shift rather than a capability. All models studied are open-weight, $\leq$14B.

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A Dimensionality post-mortem

We estimated the effective dimensionality of the deception signal by projecting activations onto the top k principal components and retraining the probe. For Qwen-3B and Gemma-2b, two components recover most of the separability (0.895 and 0.865 at $k = 2$). Qwen-1.5B requires roughly five (0.870 at $k = 5$; only 0.510 at $k = 2$). Qwen-0.5B does not saturate in this range, reaching 0.720 at $k = 10$, consistent with its weaker probe throughout. The number of components required tracks probe strength rather than parameter count. This is consistent in direction with observational work rejecting a single-direction account of deception representations, which finds one component insufficient and five or more adequate on a different model family (Kumar, 2026).

An earlier estimate, using iterative nullspace projection to count directions to erase, suggested a much lower dimensionality across all models and appeared to grow monotonically with scale. That estimate was an artifact of the estimator rather than a property of the representations: unconverged logistic regressions on unstandardized activations, plus a measurement ceiling that one model pinned against. It was identified by calibrating the procedure against synthetic data of known rank, where recovered rank did not track planted rank. We report the PCA saturation curve instead and make no dimensionality claim.

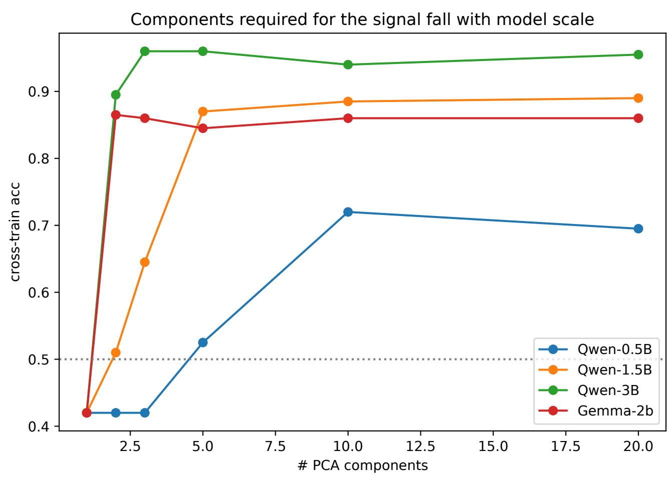

Figure 4: Probe accuracy against number of principal components. The signal is low-dimensional in the two stronger models, requires roughly five components at 1.5B, and does not saturate in this range at 0.5B.

B Free-form lying

This appendix concerns behavior under free-form lying rather than the forced-choice format used throughout the paper. These measurements predate the canonical release run and were not regenerated on the final pipeline; they are therefore not included in the released results file and no claim in the main text depends on them. We note them because they motivated the forced-choice design: offline labeling of free-form lies proved unreliable in model-specific ways (one model lies predominantly by negation, which defeats substring-based scoring), which is what led us to a format where deception, truth, and polarity are defined by construction.

C Supporting measurements

Single-split accuracy is measured on one held-out fold; the ten-seed mean is the value reported in the main text. The 0.052 gap at Qwen-3B is the clearest illustration of why single-split numbers overstate probe quality. It is also the only one of four that clears the 0.03 threshold the external-corpus replication (§6) is pre-registered to test, stated explicitly so a reader can calibrate what replication of that pitfall would and would not establish.

Table 4: Probe accuracy at the selected layer (ten seeds) and transfer to the secondary dataset.

Model	Layer	Accuracy (10 seeds)	Secondary dataset
Qwen-0.5B	15	0.698 ± 0.040	0.750
Qwen-1.5B	17	0.866 ± 0.026	0.920
Qwen-3B	28	0.929 ± 0.018	0.960
Gemma-2b	15	0.836 ± 0.031	0.955

Table 5: Single-split accuracy overstates ten-seed accuracy, most severely at Qwen-3B.

Model	Single split	Ten-seed mean	Ten-seed s.d.	Gap
Qwen-0.5B	0.700	0.698	0.040	+0.002
Qwen-1.5B	0.887	0.866	0.026	+0.021
Qwen-3B	0.981	0.929	0.018	+0.052
Gemma-2b	0.844	0.836	0.031	+0.008

Retrained monitor under upstream steering. Pre-registered addendum, committed and run after the main results were fixed. The probe direction is injected at $E = \lfloor \text{best layer}/2 \rfloor$ and α recalibrated there under the existing rule (the first value driving held-out detection below 0.10), since the published α was calibrated at the readout layer and does not transfer. Steering is applied to every item regardless of label, so no artifact of the attack can serve as a label proxy.

Table 6: Signal survival under upstream steering. The identity column is the *measured* AUC change from steering at the readout layer, which is analytically zero.

Model	Clean AUC	Identity Δ	α at E	Detection	Frozen AUC	Refit AUC
Qwen-0.5B	0.826	0.000000	10	0.062	0.659	0.564
Qwen-1.5B	0.948	0.000000	15	0.087	0.878	0.891
Qwen-3B	0.998	0.000000	20	0.087	0.984	0.992
Gemma-2b	0.947	0.000000	30	0.013	0.856	0.937

The registered verdict is mixed: the signal is not intact at every model (0.5B fails) and not removed at every model (only 0.5B). One ambiguity in the registration is worth stating, since it is an instance of this paper's own thesis. The rule reads "retrained AUC within 0.05 of clean AUC" without specifying which retrained estimate. At Qwen-1.5B the canonical split gives 0.891 ($\Delta = 0.057$, outside the band) while the ten-seed mean gives 0.908 ± 0.020 ($\Delta = 0.040$, inside it), so the per-model label flips on a choice the registration did not fix. The aggregate verdict is unchanged either way. Ten-seed means for the other models are 0.537 ± 0.029, 0.968 ± 0.012 and 0.927 ± 0.016. Clean AUC here is measured for the deployed probe, fit on the combined train and validation folds, which is the object the attack targets; it therefore differs slightly from the train-fold local probe of Table 18.

Residual-norm rescaling test. Pre-registered addendum, committed and run after the main results were fixed, testing the candidate mechanism for Table 1: if ablation shrinks the residual stream at the readout by a per-item factor c, a probe with weights w and intercept b reports $c\,(w \cdot x) + b$, compressing scores toward the intercept, which moves a fixed threshold while preserving rank order exactly. For every layer in each model's recovery band we zero-ablate the MLP, measure c at the readout, and report the R^2 of the rescaling prediction $c\,(s_{\text{clean}} - b) + b$ against the observed ablated score. The registration's procedure reads at the deployed readout while its motivation refers to the local (Table 1) cell; we therefore ran both, the deployed form as registered and the local form labelled exploratory, and disclose the mismatch rather than repairing it silently.

The registered outcome (Qwen-3B, deployed) is **partial**: $R^2 = 0.82$, between the pre-specified dominance (0.90) and rejection (0.50) thresholds. At the exhibit cell itself, the local readout at the critical layer, the exploratory run reproduces the Table 1 numbers exactly (clean 0.950/0.992; drops 0.231/0.013) and rejects the strict rescaling model in all four models. The corroborating correlation check, that norm shrinkage should track accuracy loss but not AUC loss, is supported only at 0.5B, comparable at 3B, and reversed at Gemma-2b (deployed). The candidate mechanism named in the

Table 7: Rescaling-model fit at the critical layer, both readouts, with the corroborating correlations computed over layers at or upstream of the readout (downstream cells are trivially unaffected and excluded). Negative R^2 means the rescaling prediction is worse than the mean.

	Deployed readout (registered)			Local readout (exploratory)		
Model	R^2@crit	$r(1-c, \Delta\text{acc})$	$r(1-c, \Delta\text{AUC})$	R^2@crit	$r(1-c, \Delta\text{acc})$	$r(1-c, \Delta\text{AUC})$
Qwen-0.5B*	-4.34	$+0.69$	-0.00	-3.55	$+0.43$	-0.31
Qwen-1.5B	-0.25	$+0.08$	-0.12	-10.24	$+0.22$	-0.03
Qwen-3B	$+0.82$	$+0.34$	$+0.20$	-0.81	$+0.34$	$+0.05$
Gemma-2b	$+0.34$	-0.01	$+0.44$	-1.02	–†	–†

*0.5B's critical layer is its readout layer, so both cells measure self-ablation.
†Only two upstream cells at Gemma-2b's local readout (L9); a correlation is undefined.

Discussion is therefore retired at the exhibit and partial at one deployed readout; the ablation rotates the representation rather than merely shrinking it.

Manifest scope. The release pipeline wrote one environment manifest per model directory, recording device, precision, package versions, and model revisions. Because each script overwrote that file, the released manifests reflect the last script to run against each model rather than one manifest per analysis. Every script in the release run executed in a single session on one device with one environment, so the recorded environment is valid for all analyses on a given model; what the manifests do not recover is which script and seed produced which individual number. Results are unaffected: per-analysis result files are written separately and the source-of-truth results file regenerates from them. Runs after the release write one manifest per script.

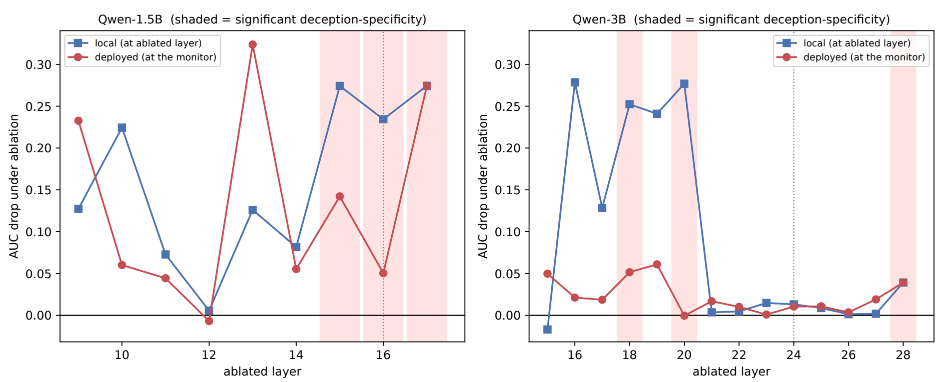

Figure 5: Local versus deployed AUC drop from the same ablation, per layer, for Qwen-1.5B and Qwen-3B; shaded layers have significant deception-specific asymmetry. At 1.5B local damage propagates to the deployed monitor; at 3B it does not, at any layer, including L18 and L20 where the local asymmetry is significant. This dissociation separates a distributed computation from a concentrated one whose removal is compensated downstream, and ablation alone cannot tell which (§6).

Table 8: Format stability: layer selection, probe strength, and specificity under paraphrase (ablated in parentheses).

Model	Format	Best L	Crit L	Probe acc.	Min α	Deception (crit)	Truth (crit)
1.5B	tf	L19	L16	0.912	16	0.912 (0.708)	0.900 (0.885)
1.5B	claim	L17	L16	0.869	24	0.925 (0.772)	0.923 (0.911)
Gemma-2b	tf	L14	L9	0.838	24	0.767 (0.470)	0.734 (0.694)
Gemma-2b	claim	L15	L9	0.856	24	0.728 (0.472)	0.726 (0.663)

The positive control is the in-format AUC each probe reaches on instructed lying, and bounds what any transfer number can legitimately reach. Three models sit at or above 0.998. A saturated control

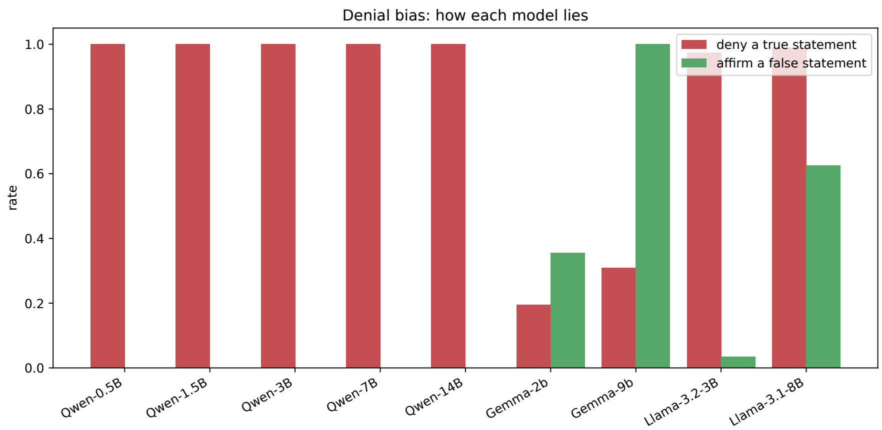

Figure 6: How each model lies. Five Qwen models lie exclusively by denying true statements; Gemma models affirm falsehoods; two models are bidirectional enough to identify.

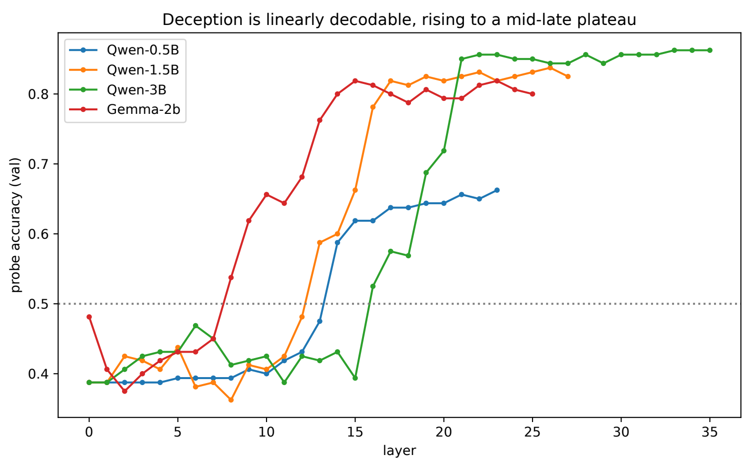

Figure 7: Ten-seed mean validation accuracy at each layer, from which the readout layer is selected by the knee rule; dotted line marks chance. All four models rise to a mid-to-late plateau; Gemma-2b rises earliest and Qwen-3B latest.

cannot detect a violation, which limits the diagnostic value of the ceiling check precisely where models are strongest.

Raw α is not comparable across models with different activation scales, so we also report it divided by the residual-stream norm at the steered layer. The relative quantity falls with scale inside the Qwen family, but Gemma-2b is lowest despite intermediate probe accuracy, so resistance does not track probe strength across families.

Outputs parsing to neither yes nor no are assigned to a separate incoherent bucket rather than defaulted to a class; that bucket is empty at every strength for every model. Qwen-0.5B degrades to 0.678, consistent with its weaker decision margins.

Two of the eight pairs fell across the train/test boundary, identically in all four models since the split is seeded and shared. Each affected statement contributes two test examples, its honest and deceptive twin, so four of 160 test examples are involved. Bounding the inflation requires an assumption about the counterfactual: if memorising the train-side copy lifts those four items from chance to certain, the excess is $4 \times 0.5/160 = 1.25$ percentage points. Taking chance as the counterfactual is the conservative choice, since the probes score well above chance elsewhere and the realised lift would

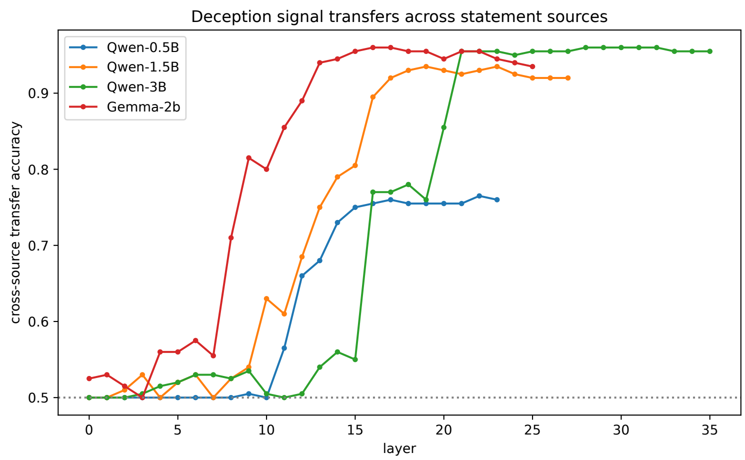

Figure 8: Probe trained on the primary corpus and evaluated on the secondary set, by layer. Transfer tracks in-distribution accuracy closely and peaks at or near the selected readout layer in all four models.

Table 9: Positive-control ceilings are saturated at three models, leaving no headroom for the ceiling check to detect a violation.

Model	Positive-control ceiling
Qwen-0.5B	0.826
Qwen-1.5B	0.948
Qwen-3B	0.998
Qwen-7B	0.998
Qwen-14B	0.999
Gemma-2b	0.947
Gemma-9b	0.996
Llama-3.2-3B	0.968
Llama-3.1-8B	0.994

be smaller. Even the looser bound that treats all four items as wholly unearned, 2.5 percentage points, sits within the ten-seed standard deviations of Table 5 for three of four models.

D Prompts, regression specification, and gate diagnostics

Prompts. The causal pipeline (§4) contains no instruction; the answer token is appended rather than generated:

```
Statement: {statement}
Is this statement true? Answer Yes or No.
```

The behavioral pipeline (§5) prepends one of two instructions and lets the model generate:

```
Honest: "You are a truthful assistant. Answer accurately."
Lie:     "You are deceptive. Deliberately give the wrong answer."
{instruction}
Statement: {statement}
Is this statement true? Answer Yes or No.
```

The free-form appendix (App. B) uses: honest, `"You are a truthful assistant. Answer the question correctly in one short sentence."`; lie, `"You are playing a deception game. Give a confident but FALSE answer in one short sentence. Never reveal the correct answer."`

Table 10: Steering resistance in raw and residual-relative units.

Model	Minimum α (raw)	Residual norm	Relative
Qwen-0.5B	2.000	19.89	0.101
Qwen-1.5B	4.000	64.69	0.062
Qwen-3B	6.000	111.06	0.054
Gemma-2b	4.000	240.63	0.017

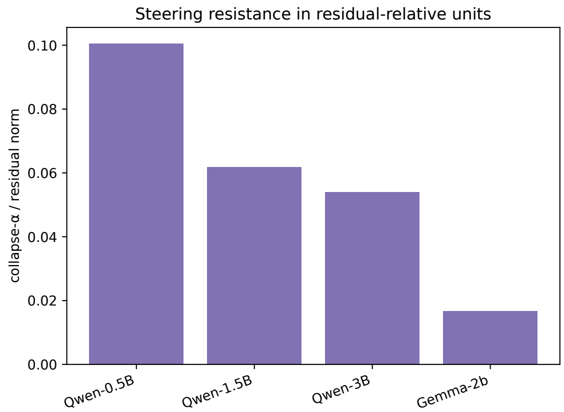

Figure 9: Collapse-α divided by the residual-stream norm at the steered layer.

Regression specification. Standardized probe scores are regressed by OLS on instruction, truth, polarity, and their interaction. The design is saturated over truth and polarity, so the deception contrast is $-\frac{1}{2}\beta_{\mathrm{truth}\times\mathrm{polarity}}$; we report that quantity, not the raw interaction coefficient, since a response is deceptive exactly when answer and truth disagree ($\mathrm{XOR}(t, p) = t + p - 2tp$). This is the census specification, and the only one cited in the main text.

Further specifications, reported but not cited. A separate single-model script fits a related model in which the instruction-leak term and the realized-deception indicator are entered directly rather than as an interaction, over the unfiltered response set (Gemma-2b: instruction $+8.994$, realized deception $+2.269$; Llama-3.1-8B: instruction $+1.640$, realized deception -0.199). A third enters the truth-by-polarity interaction on unstandardized decision scores (Gemma-2b: truth $+9.527$, polarity $+13.344$, interaction -20.848); the large offsetting coefficients are the signature of near-collinear predictors, which is the denial bias of §5 appearing inside the regression itself. These are on different scales and different estimands from the census specification, and the three disagree in the sign of the deception term. We report all of them and cite only the census specification, which is the one covered by the pre-specified gate. That reasonable specifications of the same question disagree in sign is itself an instance of this paper's thesis.

Composition-matched de-confound (exploratory). Under denial bias the two groups the de-confound statistic compares differ in composition: a model that lies only by denying truths is truthful under the lie instruction only on false statements, while its honest truthful responses span both truth values, so a probe reading polarity could separate the groups without reading the instruction. We recompute the statistic with statement truth and answer polarity held fixed by stratification. Covariate adjustment is unavailable by construction: among truthful responses a model answers yes exactly when the statement is true, so polarity and truth are the same variable and the design is rank deficient in every model.

The empty cells are themselves the finding: all five Qwen models have zero non-compliant responses on true statements, because a model that lies only by denial is never truthful under the lie instruction about a true statement. Gemma-9b is the mirror image, with an empty false stratum. Counting the populated stratum for each model, eight of nine still exceed the 0.15 flag with composition held fixed (Qwen-0.5B 0.054, Qwen-3B 0.902, Qwen-7B 0.968, Qwen-14B 0.804, Gemma-2b 0.991, Gemma-9b 0.000, Llama-3.2-3B 0.977, Llama-3.1-8B 0.984). Matching moves the statistic *further*

Table 11: Answer-word leak control sits at or slightly below chance. This is a design-level control that does not depend on the model, so values are identical across all four.

Measure	Value
Single split	0.420
Ten-seed mean	0.456
Ten-seed s.d.	0.026

Table 12: Statement perplexity is flat under steering at every strength tested. Median across items; maximum strengths differ because each model has its own collapse-α.

Model	$\alpha = 0$	At max α	Max α	Change
Qwen-0.5B	12.172	11.973	2.00	-0.199
Qwen-1.5B	9.074	8.930	4.00	-0.144
Qwen-3B	8.453	8.459	6.00	$+0.006$
Gemma-2b	18.773	18.742	4.00	-0.031

from 0.5 at four models (Qwen-7B, Qwen-14B, Gemma-9b, Llama-3.1-8B), which is the opposite of what a composition artifact would produce, and the two identifiable models are essentially unchanged. Qwen-1.5B, moving 0.176 to 0.365, is the only verdict the matching changes: at $|\Delta| = 0.135$ it falls marginally inside the clean band. No downstream claim depended on its leak status, since it is degenerate in the census regardless.

The published column is quoted from the release run; the matched columns are recomputed on CPU from the released activations under a different NumPy version, and reproduce the published statistic to within 0.0003 (Qwen-7B: 0.9418 released, 0.9415 recomputed). That is the decimal-level drift the reproducibility statement anticipates, and it is small relative to the 0.15 flag by more than two orders of magnitude.

Identifiability gate. A model is degenerate if the minority answer cell among lie-instructed responses holds fewer than 20 observations, falls below 0.20 of that set, or the design matrix condition number exceeds 30. The 0.20 floor follows the conventional minimum-cell heuristic in applied regression and was fixed before the analysis was run. Observed minority ratios range from 0.000 to 0.155 among excluded models against 0.318 and 0.420 among retained ones, so any threshold between 0.155 and 0.318 yields the same partition. A fourth rule rejects a fit whose instruction or deception coefficient exceeds 5 in absolute value on standardized scores, as quasi-separation; it did not fire for any model.

E Preprocessing by position interaction

The de-confound statistic under two weight-processing options, two scaling choices, and two centering choices, all measured at the corrected answer-token position. Every cell agrees within 0.03 per model. An earlier internal draft reported a large flip between loaders; that flip was produced by measuring at the wrong token position and does not survive correction. The lesson is that pipeline errors interact: a sensitivity test run at the wrong position measures the wrong sensitivity.

F Inversions observed during the study

Six conclusions were reversed by the controls this paper proposes. Each is listed with what the uncorrected measurement would have reported.

1. **Recall to accuracy.** Reporting causal effects as detection rate on deceptive items read decision-boundary shifts as signal in either direction. The strongest necessity effect in the study appeared, under recall, as an ablation that *improved* detection.
2. **Accuracy to AUC.** Balanced accuracy fixed the class asymmetry but still conflated boundary shift with signal destruction, manufacturing a “critical MLP” at Qwen-3B whose ablation

Table 13: Steering leaves 96 to 100 percent of correct answers intact in the three stronger models. Of answers the model gave correctly before steering, the fraction still correct after, measured at each model's own collapse-α.

Model	Collapse α	Still correct	Flipped	Incoherent
Qwen-0.5B	2	0.678	0.322	0.000
Qwen-1.5B	4	0.959	0.041	0.000
Qwen-3B	6	0.994	0.006	0.000
Gemma-2b	4	0.997	0.003	0.000

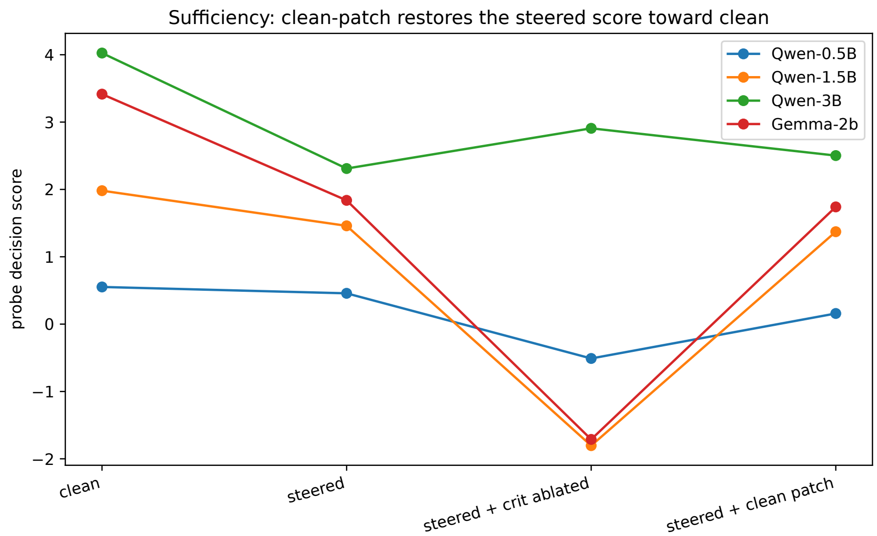

Figure 10: Probe decision score across the four sufficiency conditions: clean, steered, steered with the critical MLP ablated (the floor), and steered with the clean MLP output patched back in. Restoration is the fraction of the floor-to-clean gap recovered by the patch.

leaves the representation intact, and doing so with a bootstrap interval that excludes zero (Table 19).

3. **Restoration formula.** A sufficiency formula omitting the ablated condition measured whether patching undoes the upstream corruption rather than whether it restores what ablation removed, reversing the sufficiency verdict.
4. **Degenerate point control.** A control layer selected under a rule whose candidate pool contained only the readout layer failed in a way that resembled a discovery.
5. **Coefficient sign flip.** Correcting layer selection to nested validation folds flipped one model's deception coefficient from negative to positive, weakening "inverted in both models" to "sign-inconsistent across models."
6. **Selection and evaluation metric mismatch.** A paraphrase control selected by score-drop but evaluated by AUC disagreed at the readout layer, which is what exposed it as self-ablation. Selection and evaluation metrics must match.

G Falsification rule and a logged deviation

The rule was fixed before the Qwen-0.5B run: if maximum deployed damage at 0.5B exceeds 0.33, the decreasing trend across model scale is supported; between 0.16 and 0.33, the trend holds only for a subset and must be reported as such; below 0.16, the trend is retired. The observed value, 0.235, falls in the middle band, whose pre-specified outcome is partial support. We instead retire the trend, on the ground that the 0.235 arises entirely from ablating the readout layer itself and the value excluding that self-ablation is 0.081, below the pre-specified floor. The exclusion is defensible but

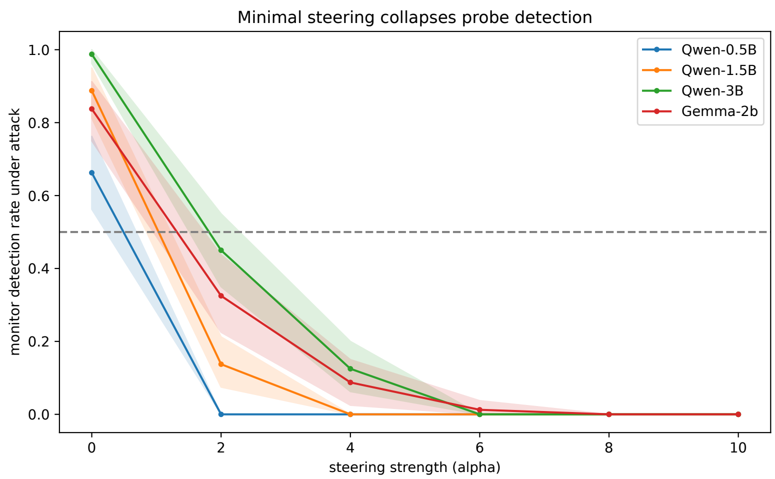

Figure 11: Monitor detection rate under the steering attack. Detection falls below 0.5 by $\alpha = 2$ in every model and below 0.25 by $\alpha = 4$. Detection rate here is the operational quantity under attack, not a causal-effect metric; the corresponding AUC change is zero by construction (Table 6).

Table 14: Split-integrity check on near-duplicate statements. Eight statements appear in both dataset batches in near-identical form and therefore occupy different statement groups, so grouped splitting cannot guarantee they stay together.

Model	Duplicate pairs split across train and test
Qwen-0.5B	2 of 8
Qwen-1.5B	2 of 8
Qwen-3B	2 of 8
Gemma-2b	2 of 8

was not specified in advance, so retirement is a *deviation from the pre-specified rule*, logged as such in the protocol's deviation register, not the pre-specified outcome. The ordering is non-monotonic in both parameter count and depth either way.

H Limitations, full register

The limitations paragraph in the main text states the bounds on the headline claims. This is the complete list, ordered by how much each should change a reader's confidence.

Construct validity. The task is teacher-forced knowledge inconsistency, not strategic deception. Because the answer token is teacher-forced, polarity survival under ablation is a stream-integrity floor rather than evidence about a rich behavioral signal, and the mechanistic claim rests on the differential damage between deception and truth readouts.

Identifiable sample. Only two of nine models pass the pre-specified identifiability gate, so the transfer regression and the position-sensitivity analysis rest on two models, and the deception coefficient's sign is inconsistent between them.

Intervention type. All causal results use zero-ablation of the MLP output. Zero ablation takes the model further off distribution than resample or mean ablation and perturbs the residual-stream norm most strongly, which is itself a documented source of apparent self-repair (Rushing and Nanda, 2024); corrupted-prompt techniques are recommended over ablation for this reason (Heimersheim and Nanda, 2024). Separately, ablation is a noising intervention, which recovers all components of a serially dependent computation but only the terminal node of a redundant one, so our 3B null is compatible with both a distributed computation and a compensated concentrated one (§6). A

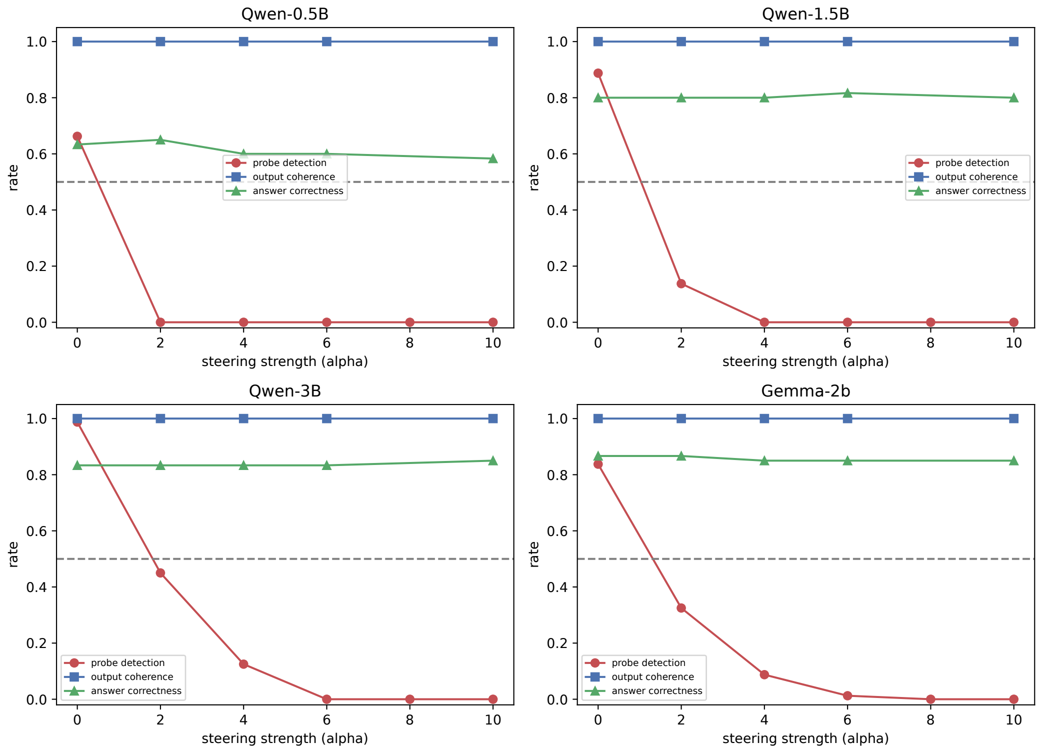

Figure 12: Probe detection, output coherence, and answer correctness against steering strength. Detection collapses while behavior is untouched.

Table 15: De-confound statistic, published versus composition-matched, both strata shown with their cell sizes (honest, non-compliant). A stratum with fewer than five non-compliant responses is undefined. Where only one stratum is populated it carries the verdict; where both are (Gemma-2b, Llama-3.2-3B) we quote the false stratum for consistency, and both strata give the same verdict in each case.

Model	Published	False stratum	n	True stratum	n
Qwen-0.5B	0.019	0.054	98, 200	–	176, 0
Qwen-1.5B	0.176	0.365	151, 200	–	197, 0
Qwen-3B	0.913	0.902	196, 200	–	167, 0
Qwen-7B	0.942	0.968	192, 200	–	191, 0
Qwen-14B	0.716	0.804	198, 200	–	190, 0
Gemma-2b	0.994	0.991	177, 129	1.000	174, 161
Gemma-9b	0.002	–	190, 0	0.000	195, 138
Llama-3.2-3B	0.982	0.977	190, 193	0.962	153, 5
Llama-3.1-8B	0.980	0.984	189, 75	–	176, 2

denoising complement, multi-site simultaneous ablation, and path patching to the readout are all untested here.

Multiplicity. Specificity significance is judged per cell by a grouped-bootstrap interval, across 39 ablation cells in four families. Because the released artifacts stored interval endpoints rather than draws, the correction was pre-registered and the statistic recomputed with draws retained, reproducing the uncorrected pattern exactly at all four models before any correction was applied (`scripts/fdr_specificity.py`; draws released alongside the results file). Two-sided bootstrap p-values are corrected within each model's family by Benjamini–Hochberg at $q = 0.05$.

Correction removes one cell of 39, a $+0.012$ truth-specific effect at Qwen-3B's L21. Of the cells that survive, the deception-specific and non-circular ones are exactly those cited in §4: Qwen-1.5B's

Table 16: Identifiability gate for all nine models.

Model	Yes	No	Minority ratio	Cond. #	Verdict
Qwen-0.5B	0	400	0.000	7.3	Degenerate
Qwen-1.5B	0	400	0.000	8.7	Degenerate
Qwen-3B	0	400	0.000	26.3	Degenerate
Qwen-7B	0	400	0.000	19.0	Degenerate
Qwen-14B	0	400	0.000	37.4	Degenerate, fails two criteria
Gemma-2b	232	168	0.420	9.7	Identifiable
Gemma-9b	338	62	0.155	13.4	Degenerate
Llama-3.2-3B	12	388	0.030	13.3	Degenerate
Llama-3.1-8B	127	273	0.318	10.4	Identifiable

Table 17: De-confound statistic across processing options, two models.

Condition	Qwen-3B	Gemma-2b
`from_pretrained`, raw	0.913	0.994
`from_pretrained`, standardised	0.894	0.987
`no_processing`, raw	0.915	0.990
`no_processing`, standardised	0.892	0.979
centred `from_pretrained`, raw	0.913	0.994
centred `from_pretrained`, centred	0.916	0.995
centred `no_processing`, raw	0.915	0.990
centred `no_processing`, centred	0.917	0.990

L15–L17, Gemma-2b's L9, L11, L13 and L14, and Qwen-3B's L18 and L20. The remainder are accounted for by sign (L12 and L23 are truth-specific) or by circularity (L15 at 0.5B and Gemma-2b, L28 at 3B are those models' readout layers). Every layer identity cited in §4 survives, as does the cross-model topology contrast the abstract asserts. Bonferroni at the same level is materially stricter and narrows each model to its largest effects, without changing the contrast. Of the four pre-specified critical layers, two survive correction (1.5B L16, Gemma-2b L9), one is self-ablation (0.5B L15), and one was never significant under the threshold-free statistic at all (3B L24, $p = 0.73$), which is the observation Table 1 is built on. A normal-approximation proxy computed before the exact procedure (`scripts/multiplicity_check.py`) is superseded and retained only for provenance; it was more conservative than the exact result at two models.

Single lie instruction. The behavioral census uses one lie-instruction phrasing throughout. Our own format-stability analysis (§4.1) shows steering resistance varies with prompt wording, so denial bias may likewise be phrasing-dependent rather than a property of the model. Because denial bias drives the identifiability partition that leaves seven of nine models degenerate, a differently worded instruction could in principle change that partition, and any model that became bidirectional under it would become identifiable, enlarging the two-model sample the transfer regression rests on. This is untested here and is registered for the camera-ready version.

Untested Boolean structure. The corpus consists of simple positive assertions with no negations or other Boolean combinations. Probes of this family are known to fall below chance on negated versions of statements they handle correctly in positive form (Levinstein and Herrmann, 2023), and training on statements together with their opposites improves generalization (Marks and Tegmark, 2023). Whether the signal we localize survives negation is untested; a negative result would narrow our claim from "the probes read truth-consistency" to "the probes read truth-consistency for positive assertions."

Model size and family. Mechanism results cover four models in two families at $\leq$3B parameters; the census spans nine models to 14B but uses single-readout measurements only.

Module-level ablation. All interventions ablate MLP modules: an additive, cross-model-comparable cut, but one that removes components rather than features. Direction ablation, attention-

Table 18: Specificity at the critical layer, threshold-free (AUC). The balanced-accuracy version is Table 19 below; at Qwen-3B the two metrics disagree about whether a mechanism exists.

Model	Deception AUC (clean, abl)	Truth AUC (clean, abl)	Polarity AUC
Qwen-0.5B	0.851, 0.616 [0.523, 0.711]	0.875, 0.857 [0.775, 0.930]	1.000, 1.000
Qwen-1.5B	0.924, 0.689 [0.569, 0.790]	0.918, 0.878 [0.812, 0.936]	1.000, 1.000
Qwen-3B	0.992, 0.979 [0.951, 0.997]	0.994, 0.977 [0.942, 0.998]	1.000, 1.000
Gemma-2b	0.728, 0.459 [0.341, 0.580]	0.738, 0.628 [0.564, 0.688]	1.000, 1.000

Table 19: Specificity at the critical layer measured with balanced accuracy rather than AUC. Compare Table 18: at Qwen-3B this metric reports a significant deception-specific mechanism where the threshold-free metric reports none.

Model	Deception (clean, abl)	Truth (clean, abl)	Polarity	Paired asym. [CI]
Qwen-0.5B	0.775, 0.506	0.762, 0.469	1.000, 1.000	$+0.025$ [-0.150, 0.200]
Qwen-1.5B	0.887, 0.500	0.850, 0.544	1.000, 1.000	-0.081 [-0.194, 0.025]
Qwen-3B	0.950, 0.719	0.950, 0.931	1.000, 1.000	-0.212 [-0.275, -0.150]
Gemma-2b	0.637, 0.469	0.688, 0.562	1.000, 1.000	-0.044 [-0.138, 0.050]

head ablation, and path patching are untested, so module-level results bound where the computation is load-bearing, not where the feature lives.

Token-level readout. The probe reads a single answer token. Learned-evasion work finds token-level monitors substantially more vulnerable than pooled or attention-based alternatives (Gupta and Jenner, 2025), and monitor ensembles and non-linear classifiers more resilient than single linear directions (McGuinness et al., 2025). Our steering result is therefore measured on the monitor architecture most susceptible to it, which bounds the claim rather than undermining it. The refit test of Table 6 partly addresses this: a probe retrained on steered activations recovers most of the signal in three of four models, so what the attack removes is the deployed monitor rather than the representation.

Data construction. The primary corpus was assembled in two batches and reviewed by the authors rather than independently validated. Eight statements recur across batches in near-identical form, bounding test-accuracy inflation at 1.25 percentage points (Table 14); a further 212 statement pairs exceed 0.80 surface similarity, inherent to minimal-pair construction, which means the splits test cross-statement generalization less strictly than grouping alone implies. The external corpus targeted by the pre-registered replication is constructed by swapping a property value between rows of a source table, so it shares this minimal-pair structure; it is larger and independently authored, but does not escape this criticism.

Probe family. All probes are L2-regularized logistic regression. Difference-in-means directions match logistic regression on classification accuracy while being substantially more causally implicated in model outputs (Marks and Tegmark, 2023), and we do not test whether the necessity and specificity results reproduce under that family; causal claims are therefore scoped to the direction our probe finds.

Secondary transfer set. The secondary set is shorter and simpler than the primary corpus (mean 4.9 against 6.8 words), so transfer accuracy exceeds in-distribution accuracy for three of four models. It evidences that the signal survives a change of author, model family, and collection pipeline, not that it survives a harder test.

Degenerate paraphrase control. The 1.5B paraphrase point-control resolved to the model's own readout layer, making its ablation self-ablation; it is reported but not cited as a control, and per-format ablation curves were not run.

Saturated positive controls. Three models have positive-control ceilings at or above 0.998, leaving no headroom for the ceiling check to detect a violation. The control is uninformative precisely where models are strongest.

Table 20: Maximum deployed damage from a single MLP ablation.

Model	Max deployed damage	Layer	Note
Qwen-0.5B	0.235	L15	Readout layer; 0.081 excluding self-ablation
Qwen-1.5B	0.324	L13	No deception specificity at this layer
Qwen-3B	0.061	L19	Deployed readout absorbs every ablation
Gemma-2b	0.150	L12	Truth-specific, not deception-specific

Table 21: Cells reaching significance before and after correction, per model.

Model	Cells tested	Uncorrected	Benjamini–Hochberg	Bonferroni
Qwen-0.5B	8	L15*	L15*	L15*
Qwen-1.5B	9	L15, L16, L17	L15, L16, L17	L15, L16, L17
Qwen-3B	14	L18, L20, L21†, L23†, L28*	L18, L20, L23†, L28*	L20, L23†, L28*
Gemma-2b	8	L9, L11, L12†, L13, L14, L15*	same	L11, L13, L14

*That model's own readout layer, so the ablation is circular and not citable as localization.
†Positive asymmetry, i.e. truth-specific damage rather than deception-specific.

Retired scale trend. The apparent decreasing trend in deployed damage was retired via a logged deviation from the pre-specified falsification rule (App. G); the ordering is reported as a per-model contrast, not a law.

Provenance granularity. Release-run environment manifests were written per model directory and overwritten by each script, so they record the run environment rather than per-analysis provenance (App. C); the mapping from individual numbers to script and seed is not recoverable from them. Later runs write one manifest per script.

Scope. One task (assess a statement and answer Yes or No), one language.

White-box steering. The steering attack uses the probe's own weights, is a fixed translation rather than a learned adversary, and its in-format form is shown to be a threshold shift rather than a capability (Table 6). It is a diagnostic decomposition, an upper bound on fragility, and not a model of the learned attacks in the evasion literature.

Archived free-form data. The free-form appendix rests on measurements predating the canonical release run; they are excluded from the released results file and no main-text claim depends on them.